

MINI ASSESSMENT

API ERROR HANDLING AND SAFE JSON ACCESS

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1

If an API returns 429, what should your program do?

FLOW OF HANDLING



Answer

A 429 response means **Too Many Requests**, usually because the API rate limit has been exceeded. The program should temporarily stop sending requests and wait before trying again. If the server provides a **Retry-After** value, it should be respected. Otherwise, use **exponential backoff**. After a reasonable number of retries, the program should stop and handle the failure gracefully.

2

If `response.json()` fails, what should your program do?

FLOW OF HANDLING



Answer

The program should catch the JSON parsing error so that the application does not crash. It should check the HTTP status and, when useful, inspect the response text for logging or debugging. The error should be handled with a clear message, and the program should either continue safely or terminate the operation in a controlled way.

3

If a JSON key is missing, why might `.get()` be safer than `[]`?

FLOW OF HANDLING



Answer

`data.get("key")` is safer when a key may be missing because it returns **None**, or another specified default value, instead of raising an exception. In contrast, `data["key"]` expects the key to exist and raises a **KeyError** when it is absent. Therefore, `.get()` is useful for optional or uncertain JSON fields.

QUICK COMPARISON

<code>data.get("key")</code>	Returns value	Returns None / default
<code>data["key"]</code>	Returns value	Raises KeyError

KEY TAKEAWAY

Reliable API programs should handle rate limits, invalid responses, and missing JSON fields explicitly to prevent unexpected failures.